Published: 18 October 2024

Citation in BibTeX format

ICCSMT 2024: 2024 5th International Conference on Computer Science and Management Technology
October 18 - 20, 2024
Guangdong, Xiamen, China

RESEARCH-ARTICLE

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Enhancing User Experience Evaluation of Graphic Art Style Games through Collaboration with Generative AI

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Abstract

With the rapid development of generative artificial intelligence (AI) technology, its application in game design is becoming increasingly widespread. This paper aims to explore how generative AI can enhance user experience by augmenting graphic art style game design. Through the study of design theories, those theories that have a strong influence on graphic art style games are identified. Next, the paper proposes a generative AI-based game design method that combines prompts confirmation, the setting of generated images, and user feedback optimization, aiming to enhance the aesthetic quality and interactivity of the game. To evaluate the effectiveness of the method, this paper designs and implements an interview method and a questionnaire to collect data and find out the user experience of generative AI-generated graphic art style games. The experimental results show that generative AI not only significantly improves the efficiency of game design, but also allows for innovative graphics and element design. The findings of this paper provide new perspectives and methods for the application of AI in future game design and also lay the foundation for further exploring the potential of generative AI in other art style games.

CCS Concepts

• **Computing methodologies** → Artificial intelligence; Computer vision; Image and video acquisition.

Keywords

Generative Artificial Intelligence, Graphic Art Styles, User Experience, Games

ACM Reference Format:
Jiajia Su and Zhongjun He. 2024. Enhancing User Experience Evaluation of Graphic Art Style Games through Collaboration with Generative AI. In *2024 5th International Conference on Computer Science and Management Technology (ICCSMT 2024), October 18–20, 2024, Xiamen, China.* ACM, New York, NY, USA, 8 pages. https://doi.org/10.1145/3708036.3708042

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ICCSMT 2024, October 18–20, 2024, Xiamen, China

ACM ISBN 979-8-4007-0999-9/24/10
https://doi.org/10.1145/3708036.3708042

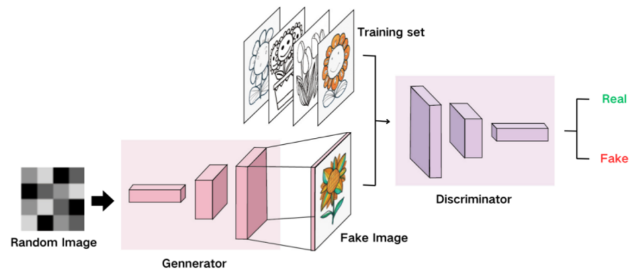

Figure 1: GANs principle.

1 Introduction

AI technology has shown great potential in image and content winnowing, especially after the introduction of Generative Adversarial Networks (GANs) and Variable Auto-Encoders (VAEs), the quality and stability of AI-generated images have been guaranteed, and various companies have introduced their models for image generation. As shown in Figure 1, GANs consist of a generator and a Discriminator. The generator attempts to convert random noise into observations that look like they were sampled from the original data set. A discriminator attempts to predict whether an observation is from a raw data set or a forgery of a generator. AIGC has been widely used in art creation, content generation, image enhancement, and image restoration.

Graphic art style games usually refer to those games that are played primarily in two-dimensional space, and come in a variety of styles, including hand-drawn art, vector art, paper-cutting art, geometric art, and many other forms. Graphic art style often creates a unique aesthetic effect through simplified or abstracted visual elements, emphasizing artistry and design. Characters, environments, and interface elements in graphic art style games tend to be flat in design, reducing three-dimensionality and emphasizing graphic outlines and shapes. This style tends to focus more on artistry and creativity, attracting players through unique visual representations. Flat art-style games require relatively few resources and skills to develop, so they are popular with independent developers. Therefore graphic art-style games have made a big splash in indie games and art games, and have gained a lot of praise from players. Generative AI has a great efficiency advantage in generating original art and game graphics for graphic art style games.

Generative AI provides new avenues for creative expression and generates new artistic paradigms that need to adhere to human-centered principles while helping designers create diverse designs

[1-2]. Akhtar and Ramkumar provide their insights on the neglect of user experience in AI design, which should build a bridge between AI-generated content and user communication, so that human intuition and AI can be seamlessly combined, emphasizing the importance of human-centredness in the process of generating content using AI [3]. Shi's study argues that AI-assisted generation of graphics is more demanding for designers, who need to have interdisciplinary knowledge, which otherwise affects the recall and accuracy of the graphics [4]. Patil focused on automated color selection and image generation from historical data in graphic design, thus demonstrating the all-encompassing impact of AI on graphic design [5]. In terms of tool research, Lin and Liu found that the AIGC tool can quickly generate a large number of creative concepts and design drafts, providing designers with more sources of inspiration [6]. Using this tool can stimulate designers' creative thinking and make it easier for them to experiment with different design styles and element combinations. In addition, Hanna studied the concept of Midjourney and analyzed its model [7]. According to Xi and Chung, Midjourney provides a new dimension for character design and improves the overall quality of animated characters [8]. Linkinen explored the challenges and benefits of generative AI in game development, which should focus on the role of the framework of game prototypes in AI-assisted game design [9]. Mi, Cao, and Li constructed a methodology for applying AIGC based on cultural elements and optimized the game character design process of AIGC in game design. Kreminski et al. generated a game case based on AI technology explored how AI technology can support storytelling and introduced design elements useful for game designers [10]. Thus, the use of AIGC in game design is well-established and widespread, and game designers are very receptive to using AIGC, but there are many issues [11].

Current research focuses on the advantages and disadvantages of AIGC technology in the game design process and its impact on graphic arts, but there is a lack of the use of AIGC to generate graphic art style game graphics and original drawings. Whether and how AIGC can be used to enhance user experience in graphic art style games is still a question that deserves further research [12].

The theme of this study is to enhance the user experience of graphic art style games by collaborating with generative AI. Specifically, this study will explore the application of generative AI techniques in the generation methodology of graphic art style games, focusing on the application of graphic art design theory in the generation process and identifying prompts about graphic art theory. Based on the cases of generating different graphic art style game screens or original game drawings, the user's experience assessment about the cases is obtained through interview method and questionnaire, so as to comprehensively understand the application effect of generative AI technology in graphic art style game design.

The significance of this study is to reveal the application potential of generative AI in graphic art style game design, and to provide new tools and methods for game designers to improve the visual appeal and user experience of games. In addition, the results of the study will provide valuable references for future AI applications in game design, contributing to the promotion of innovation and development in the game industry.

2 Research design

2.1 Process design

2.1.1 Graphic art style game screen image generation methods. Midjourney is relatively simple to use and does not require too many parameters to be adjusted, only the selection and order of prompts. Therefore more attention can be focused on the prompt rather than adjusting the parameters. Since Midjourney is based on the Discord platform, it has a user-friendly interface. As shown in Table 1, Midjourney is much simpler to use than other models. The artistic effect is more excellent, and the influence of prompts on the picture can be studied to the greatest extent.

There are several commands available for Midjourney, just type /image in the chat box followed by the prompts and suffix. It is worth noting that the command /settings need to be used with the remix model selected, this is to be able to change the prompts after they are generated, a step that helps to identify prompts later for research and testing. /show provides a command to retrieve the previously generated image, which allows you to get the seed and id of the image, which gives you the flexibility to test and adjust the prompts on the previously generated image. /describe, this function is to find the prompts of the picture, this function is very useful for prompt research afterward. /shorten, this command can streamline the prompts and filter out the prompts needed by the model.

The suffix can adjust various parameters of the generated image. The −ar is a very important parameter, and in this study, the ratio was fixed to 16:9, which is the ratio of most game images. The −s is the stylization setting, the model will be based on the set value thus determining the space for the model to play. The larger the value, the more the generated image will deviate from the prompts, resulting in a more magical image. This parameter will be used in the study of surrealist graphic art styles. However, in order to have a better study of the prompts based on the design theory, this parameter is kept as default in this study, so as to better explore how accurately the model analyzes the prompts −w is the weight setting of the uploaded image, with the value ranging from 0-2, the higher the closer it is to the original image, this parameter is used in the study of generating images from images, so this parameter will not be used in this study. The −seed suffix is used to ensure the consistency of the generated images during the optimization process. Fixing the seed number makes the generated images reproducible and this parameter is used in the optimization process. The −no is used to include the negative word.

The natural language of humans cannot be handled by machine learning, so the models are using NLP techniques to transform the input language into a language that can be understood by machine learning. Therefore, the prompt provided is very important, which affects the model's intention to understand and the graph generated. Therefore, in filling out the prompt, it should be as short as possible, as a combination of several phrases, using a combination of adjectives plus nouns. Midjourney has different functional layers. As shown in Figure 2. The base layer provides the necessary data and arithmetic support. The technology layer performs the processing of NLP and image generation. Since Midjourney has chosen Discord as its interaction platform, it needs to implement

Table 1: Suffix function Source: Author drawing.

suffix	–ar	–s	–w	–seed	–no
functionality	proportions	Stylized settings	Weight settings for uploaded images	Ensure consistency of generated images	Used to add negatives

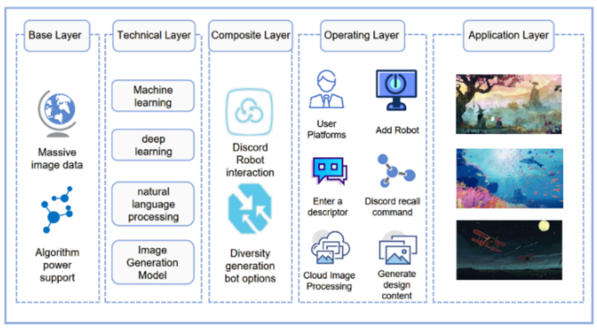

Figure 2: Midjourney platform process.

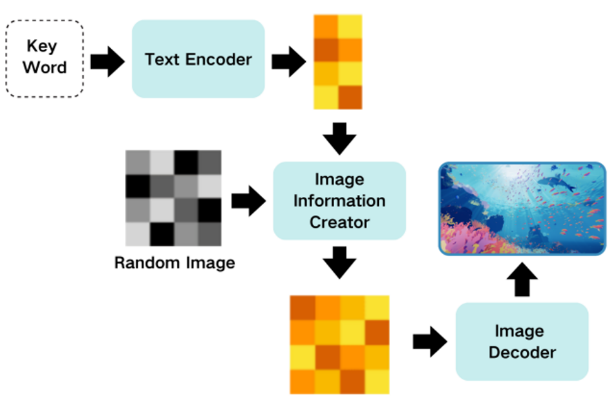

Figure 3: The process of transforming prompts into images.

user actions based on Discord. The application layer then allows for image generation.

2.1.2 Application of design theory to AI technology in the generation process. All that is needed for generative AI is a number of prompts, so it is important to translate design theory into prompts for research, which enables machine learning to better recognize and generate images. Figure 3 shows how the model converts prompts into pictures.

Artistic style is a top priority in design theory. According to the Consistency Principle, style often needs to be consistent, otherwise, it can cause visual fatigue and even interrupt the emotional connection between the user and the design work. This means that in the technical application of generative AI, there must not be too many prompts used in the overall style. According to Gestalt Theory, the overall picture needs to emphasize that the whole is greater than the sum of its parts, and human beings in perceptual vision will automatically organize all elements into a whole. Elements that are visually similar and continuous are perceived as a whole [13]. When identifying prompts, ensuring the similarity of prompts of various elements as much as possible can ensure the wholeness of the picture. According to Visual Hierarchy Theory, the arrangement and contrast of design elements direct the user's eye and attention, thus ensuring that the user can quickly find important information. This means that the order of prompts is very important, and it is better to place prompts that focus on the information as much as possible [14].

Materials create specific visual effects and are also subject to study. According to the Contrast Principle, the contrast of materials can enhance the visual effect and hierarchy. For example, a rough wooden stool can be contrasted with a smooth floor, thus adding depth and complexity to the design [15]. Materials also need to take into account the Adaptability Principle, where material selection should be adapted to the specific context of use and function. When using generative AI, identifying materials for elements is a good way to add complexity and integrity to the image.

Composition is a core concept in design and the visual arts that deals with how visual elements are arranged and organized to create a harmonious, balanced and compelling visual effect. Golden Ratio, arranging visual elements in the golden ratio, enhances the harmony of the image [16]. Leading Lines, which uses lines in a picture to guide the viewer's eyes and create depth and movement. The Golden Ratio can be used to good effect when generating architectural images, and Leading Lines can be particularly useful when guiding the player's direction [14].

Overall, style theory, material theory and composition theory all play an important role in the process of generating images with AI, in addition to this, colors, layout, and textures all affect the user's mood to some degree, which in turn affects their experience.

2.1.3 Identification of prompts of artistic style and visual elements. Identifying prompts for artistic styles and visual elements is a very important step in the design and creation process using generative AI. These prompts can help machine learning to clarify the design direction, unify the visual effects, and enhance the emotional expression of the image and the audience's empathy. There is a wide variety of art styles, each with its own unique expression and cultural background, and in this study, only graphic art styles are studied. Existing graphic art styles used in games are divided into two categories, one is traditional graphic design, which focuses on the use of lines and colors, and Grils is an example of this category. The other is geometric design, which focuses on the use of contours and shapes, and Monument Valley falls into this category. On top of the study of graphic art styles, different styles of a tour through the interview method and questionnaires to obtain the evaluation of the user experience of the prompts can obtain a more perfect picture style.

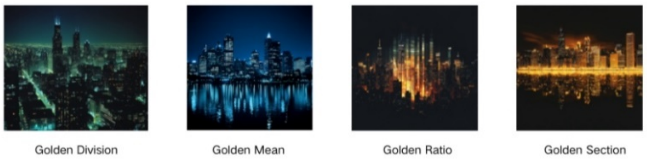

Figure 4: The effect of different "Golden Division" Prompts on the screen.

2.1.4 Application and location of prompts in the design. Generative AI has weighted distinctions for the order of prompts, so it's important to determine where these prompts should be placed. Geometric silhouettes are used as an example. Ten words are used as the prompts for the content, and the styles are placed at the very front and the very back. It can be seen that the stylistic features of geometric silhouettes are more clearly shown in the graph placed at the front. This shows that the more important prompts need to be placed at the top. As mentioned in the above study, the three theories of style, material, and composition must be in a back-and-forth order. Style has a huge impact and must be placed before the content descriptors, otherwise the results will not be particularly good. And material involves pairing with content descriptors. Composition and perspective words are flexible and can be placed before or after the content descriptors without making a big difference.

2.1.5 Prompt engineer. Generative AI is difficult to handle multiple character or subject images in the same scene. And if there is more than one subject often also leads to the lack of a subject or two subjects without hierarchical distinction. While the use of golden section prompts can effectively solve the difference between two subjects without hierarchy, the images of the game often have only one subject, so it can be solved by designing theoretical prompts. Among the extensive use of golden division terms, the model best recognizes the word golden division. This is followed by the Golden Mean. golden ratio as well as golden section are much less effective. Figure 4 shows the effect of different "Golden Division" Prompts.

The guiding line can control the visual direction of the whole picture, and you can control what object the guide line is, and the direction of the guide line. In a large number of experiments, the guide line is best paired with an object, otherwise, the guide line will appear as a separate object, potentially destroying the overall style of the image. As shown in Figure 5, when the guide line is set to tree. the AI understands this and treats the tree as a guide line. If it is not set, then it will automatically select the appropriate object to be used as the guide line.

In addition to using golden division to control the level of the main body, you can also directly control the position of the main body through the prompts. Add "in the center" after the subject. This can force the control of the subject, so as to avoid the picture clutter and no subject. Figure 6 compares the presence or absence of the keyword "in the center".

When the special prompts is particularly long, the screen details will be particularly high. When there is a conflict between the prompts before and after, the prompts will be generated according to the former. Thanks to the progress of the trained model, the details of the description are represented very accurately. As can be seen, of the three plots in Figure 7, the second plot with conflicting prompts is not very different from the first. But after removing them, the third figure is particularly different from the first.

Figure 5: Comparison of adding "guide line" and specifying guide line as tree.

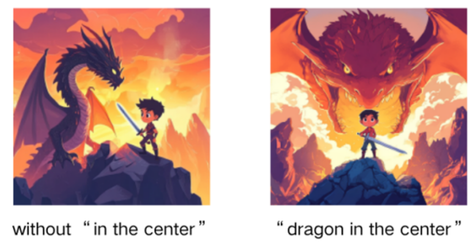

Figure 6: Comparison with and without "in the center".

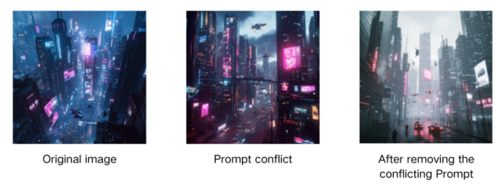

Figure 7: Comparison of When Prompt Conflicts.

2.1.6 Comparison of different models. Table 2 shows the basic information of the three mainstream AI-generated image models. As shown in Figure 8, three different models are used to generate images using the same prompt, with all the parameters as default. the quality of the images generated by Midjourney is much better than that of the other two models. Especially for the processing of "gameplay footage", the image generated by Midjourney is very much like the gameplay footage. The other two models have some processing but are not as fine as Midjourney. In terms of "golden division", it is obvious that Midjourney is more in line with the principle, and DALL-E3 also achieves a certain effect, while Stable Diffusion is completely symmetrical in the center. On the "guide line", all three models have some effect, but Midjourney is more in line with this effect. Therefore, based on the quality of Midjourney's images and its accurate handling of the design theory prompt, Midjourney will be used in future research.

2.1.7 Example of generating game screen images with different graphic art styles. As shown in Figure 9, 1a with all the design

Table 2: Comparison of different models.

Platform name	Development Company	It is open source	Highlighting Advantages	Image Processing
Midjourney	Midjourney	NO	Simple interaction, high artistry	Cloud
Stable Diffusion	Stability AI	YES	Abundant controllable plug-ins and stable quality	Local
DALL-E 3	Open AI	NO	Strong textual semantic understanding	Cloud

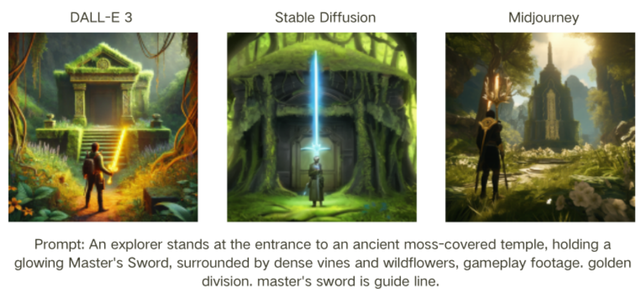

Figure 8: Comparison of images generated by the three models.

theory prompts removed, the picture is more exquisite but lacks the effect and design aesthetics of a game. And the style of the generated picture is not certain to achieve the desired effect.1b After writing through the prompts of the picture style, the precise art style of the picture is obtained. But it still looks like a poster rather than the original art of the game. When the wide-angle lens of the game is added, the overall picture is more like the game picture. 1c is the final generated picture, and the composition and the quality of the picture are somewhat improved after adding the design theory prompter.

2a In the case of only describing the picture instead of using the design theory prompts, it can be seen that thanks to the model the picture exquisiteness is guaranteed, but the details are too trivial and unfocused.2b When the prompts of style and game picture are added, the style is more unified and more in line with the original game picture.2c After adding the final design theory prompts again, the overall picture has a focus and a clear hierarchy, which is more in line with the public's aesthetics.

3a In the case of description only, the picture is rough and unfocused. The contrast between light and dark is poor. 3b with the addition of the style has increased detail and the contrast between light and dark has improved significantly. However, the composition is not good enough to find the focus of the picture, which is not at all in line with the requirements of a game original drawing. 3c After adding the golden section prompts, the overall contrast between the light and dark of the picture is strong, the leading line is prominent, and the golden section is used to emphasize the theme.

Six different styles of game screen images were generated by combining the prompts obtained from the above study.

Figure 10(a): whimsical landscape with a cartoon art style, perspective effect,wide shot of the game scene ,a pink sky with clouds,distant peak,a big tree sits on the left side,magician is casting spells, young apprentice is watching, baguette is a visual guide,golden ratio –ar 16:9. Figure 10(b): simple geometry, game scene,central composition, overlooking the camera,pink game environment with smooth tall columns,simple shapes, and simple lines of minimal detail,color-specific person stands on top or in between them,soft lighting and shadows, low contrast, matte colors and background–ar 16:9. Figure 10(c): style of Pixar,game scene,a vibrant and colorful coral reef with fish swimming around, the scene is bathed in bright sunlight that filters through blue water, creating a sense of depth and enchantment. this illustration captures the essence of underwater exploration and adventure. –ar 16:9. Figure

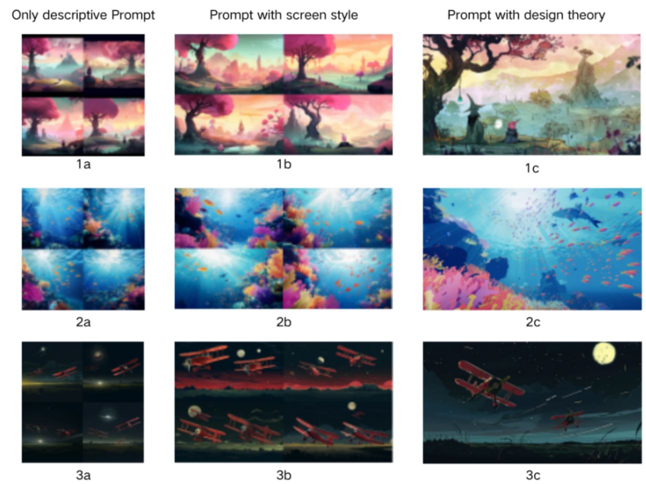

Figure 9: Comparison of Prompt with Description, Prompt with Style and Prompt with Design Theory

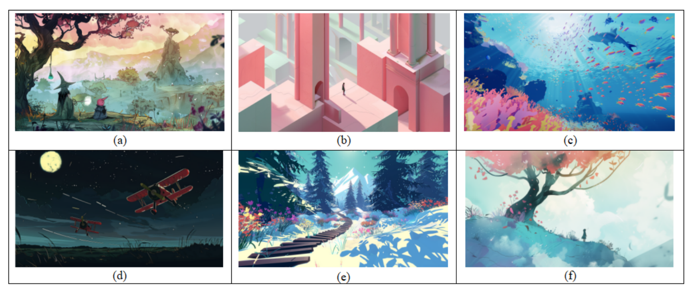

Figure 10: Case Screenshot.

10(d): cartoon-style illustration, golden section, two biplanes flying over the field at night, shooting with their guns,the sky is full of the moon and stars,one plane has red wings,while another one flies in front of it,in the style of comic illustrations. –ar 16:9. Figure 10(e): a snowy mountain path leading into the distance, surrounded by dense forests and colorful flowers on both sides of it, in the style of a flat illustration, 2D game art style. the background is a blue sky with white clouds on a sunny day. it has a cartoonish and fantasy world style with bright colors and a high resolution. there are forest trees like pine trees and snowflakes but no characters in front or shadows. –ar 16:9. Figure 10(f): sical illustration of an old tree with red leaves, standing on the edge of a hill overlooking a vast blue sky and white clouds. A small girl in pastel stands under its branches, looking up at her colorful petals falling gently around them. The scene is captured from behind as if she's walking away, creating depth and perspective through a simple color palette of pink and teal tones. –ar 16:9.

Six drawings were generated and screened according to the design process. Figure 10(a) and Figure 10(f) are in watercolor hand-painted graphic style using visual hierarchy theory. The visual near view is large but occupies a smaller position, and things in the distance occupy a large position, resulting in a perspective effect. The inconsistency of the prompts perspective description will appear in different picture effects. Figure drawing sends a warm feeling, while Figure 10(f) drawing is in the back of the distant perspective, it gives a lonely, the protagonist perseverance feeling.

As shown in Figure 10(b), this drawing adopts the flat style of minimalist vector graphics. Using the design principle of central composition and the principle of contrast, the viewer's perspective is focused on the center of the image. The contrast between the tiny characters and the huge game world makes the picture more shocking; Figure 10(c) adopts the graphic style of cartoon caricature and uses the principle of contrastivity to contrast the blue-colored ocean world with the dazzling corals and colorful small fishes at the bottom of the sea, which adorns the blue ocean more beautifully. Adopting the principle of adaptability and Gestalt, because it is a water world, so all the visual effects should obey the effect under the sea surface.

Figure 10(d) is the graphic style of cartoon comics. The visual hierarchy theory is applied. The visual goes from the red plane to the black night, showing the coldness and blood of war. The prompt is the golden section theory, which makes the relationship between the subject matter and the picture more harmonious and beautiful; Figure 10(e) is in 2D game art style, using the theory of visual hierarchy theory and the law of the center composition, extending the perspective from the very front of the picture to the center of the picture. The scene depicts a quiet path leading to a snowy mountain, with dense forests and flowers growing on both sides.

2.2 Questionnaire and interview method

2.2.1 Questionnaire design and implementation. The goal of this questionnaire was to investigate the acceptability and usability of AI-generated graphic art style game graphics and original game art for different groups. In order to achieve this goal, a series of detailed questions were designed and implemented to obtain respondents' evaluation scores of these images. The questionnaire was distributed online to a total of 57 respondents. The questionnaire focused on two main areas: whether the respondents could recognize whether the images were generated by AI or not, and the ratings of these images as game graphics and original game art. These ratings helped to understand the aesthetic and utility ratings of AI-generated images by different groups.

During the survey process, to ensure that the results were comprehensive and representative, each questionnaire contained specific rating items, and respondents provided detailed ratings for each image. Through this data, it was possible to distill users' actual feelings about AI-generated art and suggestions for improvement. This information is also vital for designers to help them improve the techniques and styles of AI-generated art to make it more in line with users' aesthetics and needs. And by improving the quality and appeal of game graphics and original art, designers can create more popular and visually appealing game titles.

2.2.2 Interview subjects and methodology. Through a series of well-designed questions, this interview comprehensively investigated users' experiences and perceptions of AI-assisted game screen design, including feedback in a variety of areas such as aesthetics, functionality, acceptance, and willingness to recommend. To ensure the comprehensiveness and reliability of the data, each interview covered a different group of users, including design students, emerging artists, and gamers. Each interview provided insight into the specific needs and experiences of the respondents. During the course of the interviews, special attention was paid to the benefits and challenges of AI technology in graphic art style game design. A wealth of user feedback and suggestions were gathered through in-depth dialogue, with suggestions covering not only the current performance of AI-assisted design, but also user expectations and specific suggestions for future improvements. In addition, based on a detailed analysis and collation of the interview data, key insights and trends in AI-generated graphic art-style game prototypes were distilled. These insights provide valuable references and improvement directions for future AI-assisted game design.

3 Analysis of results

3.1 *User feedback and suggestions*

Through the questionnaire survey, it was found that the percentage of those who could identify AI images was 64.7% (Figure 11), which means that the majority of people were able to identify whether the images were made by AI or not. As for the reason for successful recognition, most of the interviewees thought that the reason was because the elements of the picture were too strange. Although the difference in results between different image styles was relatively small, the overall result was that the minimalist silhouette style was difficult to identify as AI-generated, followed by the hand-drawn style, while the 3D style was very easy to identify. The average score for the generated images as original game art and game graphics was 7.785 (Figure 12), which indicates that most of the interviewees were very satisfied with the quality of the game graphics, with the only shortcoming being that the elements of the graphics were too strange and not uniform in style.

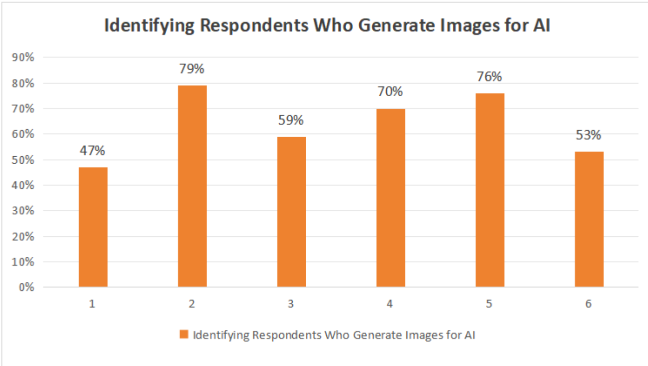

Figure 11: Identifying respondents who generate images for AI.

Figure 12: The mean value of each image.

The interview method is used to understand the views and feedback of different groups on AI-assisted graphic art style game design. AI mapping has higher efficiency and better innovativeness compared to manual mapping, which can generate wild ideas. These special ideas added to the game will bring players a unique experience. However, the lack of detail in the graphics of AI mapping is a big difference from the needs of the game which should focus on the sense of experience. The game's graphics should meet the player's needs, and the design of the game's mechanics is more important than the game's graphics.

The main problem in the lack of detail in the image of AI mapping is that the overall image is only the main body, the details of the background are not portrayed properly, and it lacks clarity and sophistication, thus affecting the overall visual effect. As images with rich details can stimulate users' emotional resonance and imagination, images with insufficient details are difficult for users to resonate with, thus reducing the infectiousness and attractiveness of the artwork.

In addition, the most important aspect of a game is the sense of experience and game mechanics. Game mechanics/core gameplay is the foundation of the game, which determines whether players will be interested in the game and play it for a long time. No matter how beautiful the graphics are, if the core gameplay is not engaging, players will quickly lose interest. A compelling story and worldview can enhance the immersion of the game, making players more engaged in the game and increasing their experience. And great game mechanics and sense of experience can inspire emotional resonance in players, making them care more about the characters and environment in the game. If the characters in the game lack details, such as facial expressions, clothing textures and movement details, it will make the characters look hard and unrealistic, thus affecting the player's sense of immersion and game experience. A game scene that lacks details, such as the foliage of trees, the dynamic effects of water flow, and the subtle features of buildings, will make the whole scene look bland and uninteresting, reducing the player's desire to explore. The design of game graphics really needs to be tailored to the needs of the player to ensure that their experience is enjoyable and satisfying. Whilst the game mechanics and core gameplay are crucial, the quality and style of the graphics are just as important as they directly affect the player's first impression and immersion.

3.2 *User feedback and suggestions*

The questionnaire allowed respondents to score the case screenshots in Figure 9 individually to assess the aesthetics and functionality of the images in terms of analyzing the aesthetic principles. As shown in Table 3, the scores were significantly improved when design theory was added to the prompts, Figure 9 (a) performed best in terms of image composition and color tone, and Figure 9(c) was significantly improved in terms of image composition and innovativeness, and Figure 9 (d) was significantly improved in terms of image composition and image harmony. When design theory prompts are added, the scores of each of the images generated by generative AI will increase significantly.

3.3 *Verification and improvement of design theory*

To address the lack of detail, more detail can be expressed by adjusting the resolution of the generated image. Adjust the prompt and use more descriptive language and clear instructions to define more precise details. Therefore, the prompts must not only take into account the style material and composition, but also the details of the main content must be described in detail. According to the principle of consistency, the description of details should be clear, specific and coherent to ensure that the resulting image has consistent detail and style in all aspects and does not mix incompatible styles. For example, if the description is of an old medieval castle, then all details should be consistent with the medieval style. In addition, consistent visual elements and themes need to be maintained in the description. For example, if describing a natural landscape, make sure that all details fit the theme of the natural environment. According to the visual hierarchy theory, describing different parts of an image in layers to ensure that each level of detail is fully described helps Midjourney to understand and generate more complex and nuanced images.

To address the issue of game graphics design, the target audience of the game needs to be identified thus finding the preferences and expectations of the players. Market research can be done to understand the needs and preferences of the target audience for the game graphics. Ensure that the graphic style of the game is consistent with the theme and emotional tone of the game. As well

Table 3: Questionnaire results with images generated by Design Theory Prompt and original images.

	a	A:Prompt without design theory	c	C:Prompt without design theory	d	D:Prompt without design theory
Picture composition	9.12	5.26	6.16	4.32	8.41	6.21
Standardize and diversity	7.15	6.12	7.65	6.55	6.32	5.65
Picture Harmony	8.96	7.32	7.46	7.65	7.45	7.65
Color and Tone	9.35	8.55	8.72	8.92	7.76	7.56
Innovativeness	7.26	6.65	7.31	6.96	6.45	7.25

as, maintaining visual consistency of all elements within the game, including character design, scene layout, props and user interface, which helps to enhance player immersion. Rich environmental details can enhance the realism and immersion of the game, using a sensible color palette to create atmosphere and emotion. Therefore, before generating images, the worldview and storyline of the game should be determined, and based on this, the details should be described in detail to ensure that the style is consistent with the story and worldview.

4 Conclusion

The research significance of this thesis is that it explores the application effect and user experience evaluation of generative AI technology in graphic art style game design, which fills the research gap in the related field. This research provides new design ideas and tool support for game designers, which helps to promote the innovation and development of game design.

In summary, this study verifies the importance of details and consistency and proposes improvement measures by analyzing the effect of graphic art style game screen design. In future game design, it is necessary to comprehensively consider the richness of details, user experience, game mechanics and the coordination and unity of the screen style, in order to create high-quality game graphics and original drawings that are both visually appealing and have a sense of immersion and attraction. This study is also a little bit insufficient in that it only focuses on Midjourney and ignores other big models and algorithms. Different models should be explored in detail in order to fully demonstrate the problems associated with generative AI-generated graphic art style game graphics.

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